
Research Article

Kinetic Model Development for Accelerated Stability Studies

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Abstract. Accelerated stability coupled with modeling to predict the stability of compounds, blends, and products at long-term storage conditions provides significant benefits in science-based decision-making throughout drug substance and drug product development. The study can often be completed, including data analysis in the space of three working weeks, and the information gathered and learning made in this time period can rival years of traditional analysis. The speed of the studies allows an earlier assessment of risk to quality enabling appropriate risk mitigation strategies to be implemented in a timely manner. The scientific foundation is based upon Arrhenius kinetic equations that can be linear or nonlinear in time, and can be based upon water vapor pressure or liquid water activity (relative humidity). A variety of kinetic models are evaluated, and the best model is chosen based upon both Bayesian information criteria and an automated assessment of kinetic model parameters fitting within acceptable ranges. Confidence intervals are estimated based upon a bootstrapping approach. Moisture vapor transmission rate models are applied on top of the resulting kinetic models in order to simulate different packaging types and the use of desiccant. The kinetic models are integrated with the prediction of packaging humidity over time to create a long-term prediction of impurities and other phenomena. The resulting models have been shown to be useful for not only the prediction of drug product impurities in long-term storage but other physical phenomena as well such as hydrate development and solvate loss.

KEY WORDS: accelerated stability; kinetics; stability; stability modeling.

INTRODUCTION

Accelerated stability studies observe drug product or drug substance stability at high temperatures and use modeling in order to predict the stability at lower temperatures under long-term storage conditions. This approach provides significant benefits in science-based decision-making throughout drug substance and drug product development.

Accelerated stability has been used throughout the pharmaceutical industry and has had significant exposure to regulatory agencies across the globe (1). The concept is proposed as an improvement on current working practices for assigning retest periods for intermediates and starting materials, as well as providing vital understanding of a product's stability. In contrast to typical accelerated stability or forced degradation studies, these accelerated stability studies are typically carried out over 2–3 weeks at temperatures up to 80°C and relative humidity up to 90%. The study can often be completed, including data analysis in the space of three to five working weeks, and the information gathered and product

knowledge generated in this time period can rival years of traditional analysis. The speed of the studies allows an earlier assessment of risk to product quality, thus enabling appropriate risk mitigation strategies to be implemented in a timely manner during development.

Samples are stored in a variety of temperature and humidity conditions determined by experimental design. Changes in the attribute of interest (impurity level, drug substance assay, dissolution performance) are monitored over the course of the study by periodic sampling, and the output modeled to provide a prediction of the long-term stability.

The scientific foundation for modeling the resulting data is based on Arrhenius kinetic equations to create best fit predictions of the degradation of a substance as affected by temperature and humidity. The modeling works by deconvoluting the effect of temperature, time, and humidity on the rate of degradation and does not assume a linear relationship to any of these factors. This approach results in more accurate modeling than traditional accelerated studies where a common increase in degradation rate with temperature is assumed for all products. In a regulatory context, it is usually assumed that a period of time at 40°C/75% relative humidity (RH) is equivalent to double that time at a “long-term” storage condition.

Moisture vapor transmission rate (MVTR) models are applied in conjunction with the kinetic models in order to

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simulate different packaging types, further increasing the usefulness of the approach. Control strategies can be developed by evaluating the effect of packaging and desiccant. This approach enables the same set of stability data to be utilized for more than one packaging type and for more than one “climatic zone.” This reduces resource and speeds up development time where different variations may need to be set down on long-term stability in order for data to be generated to support a product decision.

Currently, the most widely published approach for modeling drug product degradation based on accelerated stability data is that described by Waterman (2,3). This is based on a modified Arrhenius equation which assumes an exponential relationship between reaction rate and environmental RH.

Typically, the impact of concentration of reactants on a reaction rate is captured *via* a kinetic (rate) equation, where the concentrations, or more strictly the activities, of reactants are raised to some power, also referred to as the order of the reaction, rather than as an exponent. Adami and Waterman (3) assert that the need for an exponential relationship is based upon observation that an exponential fit linearized the reaction rate for two prior studies (4,5). Based on these two studies, it was hypothesized that water activity is best modeled as an exponential function of RH.

At GSK, we have found that water is best treated as a reactant within a kinetic model rather than as an exponential parameter. The evidence of this is that humidity-sensitive products exposed to significant desiccation have degradation rates slowed down as predicted by the concentration of water in this system raised to a power, not an exponent of RH. Since an exponent of near zero cannot approach zero, an exponential equation *versus* RH overestimates the resulting reaction rates near zero concentration of water. This is an important distinction, because for products that could successfully be packaged with a desiccant as predicted by a kinetic equation, an equation that assumes an exponential RH relationship would send these products toward more expensive refrigerated storage. We have also observed slightly improved models with different equation terms that also consider that water could be reacting as a vapor or as a liquid.

There is an “isoconversion approach” described in literature (2) and popularized as an alternate approach to fitting time series data. A model is used to predict time until failure which avoids the complexity of fitting kinetic model parameters that describe how the rate of degradation varies over time. Each sample within the experimental design is driven to failure, typically the specification limit for a given impurity, and then the time (*t*) to failure point alpha is predicted.

Using an isoconversion approach, every experimental condition within an accelerated stability study should be driven to failure; otherwise, the time to failure needs to be extrapolated for the milder conditions, leading to unknown extrapolation errors for each condition. In practicality, this limits the range of temperature and humidity that can be studied in a given time limit to the more extreme conditions that drive impurity formation to failure within the time available for study. This leads to a strategy of limiting the temperature explored to higher values, resulting in a model that will have higher uncertainty for the activation energy that has to be extrapolated further to long-term storage

conditions. Higher temperatures needed for these types of studies also risk promoting thermal degradation reactions or changes in phase that are not practically relevant at lower temperatures.

Another disadvantage of the isoconversion approach is that the humidity increases substantially with time in packaging in real storage conditions; therefore, an estimated mean kinetic humidity has to be calculated to enter into the model and predict time to failure. This results in an unknown error of approximation for the mean kinetic relative humidity for the time period. In contrast, a kinetic equation can be integrated with time as the humidity varies each day and avoids the need to make this approximation.

Our desire was to avoid the use of the isoconversion approach and fit kinetic models directly to the attribute *versus* time data instead, as this avoids the concerns and complexities listed above, resulting in a strategy where experiments can be run for a fixed amount of time, and makes full use of the information available for impurity amount *versus* time. It also allows projecting reactions that are not strongly accelerated under more forcing conditions and thus cannot be tested to failure in a reasonable amount of time. We have observed examples of impurities that are barely observed above the limit of detection under accelerated conditions that turn out to be the most problematic impurity at lower temperatures due to small activation energies. An example of this is shown in this paper in the section comparing fitted models to long-term trends for product 81. Fitting a kinetic model requires fitting up to one additional parameter compared to an isoconversion approach, but adds the complexity that there may be multiple types of curvature with respect to time. As a result, different kinetic equations will need to be evaluated. Since a typical accelerated stability study will have 19 to 25 data points, the use of a kinetic approach leads to 16 to 22 degrees of freedom for fitting error which should typically allow good capability for discrimination between the kinetic model types discussed in this paper.

Khawam and Flanagan (6) provide an excellent review of solid-state kinetics in the pharmaceutical industry from a first principles basis, and it was a paper accompanied with more in-depth PhD thesis (7). This review and accompanying PhD thesis proved particularly useful in the development of our approach.

THE ROLE OF WATER IN REACTION KINETICS

Under typical storage conditions, water exists as either a liquid or a vapor and can interact readily with solids in either phase. Formation of an impurity could be due to adsorption and reaction of water vapor at a boundary layer with the atmosphere, or due to interaction with liquid water adsorbed on the solid surface, or liquid absorbed within the bulk of the reacting material. It could potentially be important to make the distinction as to the source of water (liquid or vapor) as it has implications on whether RH or water vapor pressure (VP) is the most appropriate term for use in modeling. The kinetics can potentially be different depending upon the phase of a reactant. Selection of the wrong model term (RH or VP) could potentially result in a kinetic model that cannot be significantly extrapolated outside the temperatures studied in the experiment without concern about systematic errors if the kinetics are sensitive to moisture activity.

Kinetic Model Development for Accelerated Stability Studies

Water activity of a solid has a direct relationship with RH when at equilibrium, and so relative humidity is an appropriate term to use in kinetic models when liquid water activity is affecting the outcome. There are numerous examples of reactions and physical changes that are documented as being RH or water activity dependent, suggestive of interactions with water in dissolved liquid state (8–12). There are also physical transformations and reactions with rates that are reported to be proportional to water vapor pressure, suggestive of direct interaction between the solid and water in the vapor state (13–16).

Based on the studies conducted in GSK to date, we have noted that solid-state phase changes, desolvation, hydrolysis, and secondary crystallization are typically vapor pressure dependent.

Aspirin is a classic example often cited where the hydrolysis rate is directly proportional to the water vapor pressure in the environment (17). The US Pharmacopoeia (18) states the following regarding hydrolysis:

Esters and β -lactams are the chemical bonds that are most likely to hydrolyze in the presence of water. For example, the acetyl ester in aspirin is hydrolyzed to acetic acid and salicylic acid in the presence of moisture, but in a dry environment the hydrolysis of aspirin is negligible. The aspirin hydrolysis rate increases in direct proportion to the water vapor pressure in an environment.

In light of this, there is a need to consider the possibility of either water vapor partial pressure, or relative humidity when developing a kinetic model from accelerated stability studies. Wrong selection raises the possibility that if the degradation rate is particularly humidity dependent ($|N_1| > 1$), and the extent of extrapolation with respect to temperature is large (>30 degrees), a noticeable systematic error will occur because vapor pressure does not increase in a precise exponential fashion with temperature, and thus cannot be fully taken into account in an activation energy term within a fitted kinetic equation. This creates some manageable complexity to consider when designing an appropriate experiment for accelerated stability, as inappropriate experimental design will result in confounding and increased error in the estimation of model parameters. Experimental design for accelerated stability studies is discussed later in this paper.

KINETIC MODEL DEVELOPMENT

Change in reaction conversion (called alpha in this analysis) as a solid active pharmaceutical ingredient (API) reacts with water that is typically related to the “activity” of the reactants, raised to a power (N_1), which is referred to as the order of reaction (19) as follows:

$$\frac{d\alpha}{dt} = k \cdot [\text{H}_2\text{O}_{\text{activity}}]^{N_1} \quad (1)$$

Strictly, Eq. (1) should also include the activity of the API raised to a power, as the above equation assumes a fruitful molecular encounter with water to generate the

degradation impurity; however, since we are typically only interested in impurity formation to 1% and at most 10%, we can assume that the change in API content with respect to time is close to zero; thus, this term can be lumped into the observed rate constant k . Degradation reactions that do not involve the direct reaction of API with water are also expected to follow the rate equation above since, in many cases, these reactions are also strongly impacted by the water activity, *e.g.*, higher rate of degradation with increased water activity due to increased molecular mobility or lower rate of degradation with increased water activity where water itself is a by-product of the degradation reaction for example in the formation of an ester or amide linkage in a degradant.

In the case of liquid water adsorbed onto the surface of a material, water activity is the ratio of vapor pressure of water in the product to the saturation pressure of water vapor at the same temperature. For solids at equilibrium with the environment, the water activity is correlated with the relative humidity. Thus, for a reaction occurring due to liquid water, the generic rate Eq. (1) becomes (2).

$$\frac{d\alpha}{dt} = k \cdot [\text{RH}]^{N_1} \quad (2)$$

Where RH is the relative humidity expressed as a fraction, and N_1 is a fitted power term.

For a reaction occurring as a solid interacting with a vapor, the water activity of the reactant is equal to the concentration of water molecules in the atmosphere. The concentration of a vapor is equal to its partial pressure, which is defined as the vapor pressure divided by system pressure. Equation (1) for a vapor phase interaction is then defined as (3):

$$\frac{d\alpha}{dt} = k \cdot [\text{VP}]^{N_1} \quad (3)$$

Where VP is the vapor pressure of water divided by atmospheric pressure (typically 1 atmosphere) and is equivalent to the gas phase concentration of water vapor. Since vapor pressure follows an approximate exponential curve with temperature, distinguishing between liquid *versus* vapor phase-driven reaction kinetics may not always be critical, but if the exponent of N_1 is significantly greater than 1, it can in some cases add error of extrapolation on order of 10% to 15% if the predominant reacting phase is not correctly identified. This is because while vapor pressure follows an approximate exponential relationship with temperature and thus a difference could be approximately wrapped into an activation energy term, the relationship of vapor pressure to temperature is not exactly exponential. It is therefore useful to test the fit of both vapor pressure and RH as alternate hypotheses to examine which one best fits the data at hand, and to also consider the implications if either the VP or RH hypothesis were true. For most reactions, the difference in extrapolation may not be observable in the data.

The k term in these equations is an Arrhenius factor relating an exponential relationship between temperature T ,

the activation energy of the reaction E_a and the gas constant R . Expanding on the Arrhenius term for Eq. (3) yields (4)

$$\frac{d\alpha}{dt} = K_1 \cdot \exp\left(\frac{-E_a}{R \cdot T}\right) \cdot RH^{N_1} \quad (4)$$

Where K_1 is a pre-exponential term called the “frequency factor.” For simplicity of nomenclature and units in the models described in this report, the activation energy divided by the gas constant (E_a/R) will be simplified to a fitted parameter K_2 as shown in (5).

$$\frac{d\alpha}{dt} = K_1 \cdot \exp\left(\frac{-K_2}{T}\right) \cdot RH^{N_1} \quad (5)$$

Fitting Eq. (5) can be challenging as K_1 and K_2 are correlated in an exponential relationship. A way of reducing this exponential correlation between K_1 and K_2 is to scale the exponent to a value of 1.0 at the average temperature in the study (T_{ref}) by modifying Eq. (5) to create Eq. (6).

$$\frac{d\alpha}{dt} = K_1 \exp\left(\frac{-K_2}{T} + \frac{K_2}{T_{ref}}\right) \cdot RH^{N_1} \quad (6)$$

Scaling the exponent by the use of a reference temperature (T_{ref}) has no impact on the activation energy calculated but reduces the correlation between K_1 and K_2 , allowing for more robust estimate of these parameters.

What is actually fitted to the experimental data is the integrated form of the kinetic models, as the final impurity value at time “ t ” is what is observed. Integrating (6) with respect to time yields (7)

$$\alpha = K_1 \cdot \exp\left(\frac{-K_2}{T + 273.15} + \frac{K_2}{T_{ref}}\right) \cdot (H)^{N_1} \cdot \Delta t + C \quad (7)$$

Where C is the value of α at time zero, and H is the humidity expressed as either RH or vapor pressure. The type of humidity dependence will need to be tested within the experimental analysis as two alternate models.

Equation (7) is the simplest model form, noted to be linear with respect to time, and requires three fitted

parameters K_1 , K_2 , and N_1 . The initial average impurity content C should be measured and known.

In many cases, the rates of many solid-state degradation reactions will not remain constant with time. There are a large number of potential kinetic model forms that are nonlinear in time and could be considered for possible use. For example, Khawam and Flanagan (6) lists 17 possible models that could be selected, and the different model forms describe reactions that speed up with time, are linear with time, or slow down with time. However, with the assumption of low extent of conversion, the 17 models can be reduced to five categories that can be adequately fit most of the time with a single model for each reaction category.

The five types of kinetic models selected for the accelerated stability modeling (ASM) platform and their associated typical curves are summarized in Fig. 1 and Table I.

It should be noted that the H term in the models is either the dimensionless relative humidity or the dimensionless vapor pressure of water divided by the atmosphere pressure. Both RH and VP versions of each model are developed.

For simplicity, all fitted parameters are referred to as K_1 , K_2 , K_3 and fitted power terms as N_1 , N_2 . The terms in the models do have scientific meaning as follows:

α	is the impurity level, assay loss, or measure of extent of physical transformation.
H	is either the vapor pressure measured in atmospheres or the relative humidity (dimensionless).
K_1	is the fitted frequency factor at the reference temperature.
K_2	the fitted activation energy divided by R , or E_a/R , with units of K^{-1} .
K_3	is the asymptotic limit of degradant at $t = \infty$ due to limiting reagent.
T	is temperature in $^{\circ}C$.
T_{ref}	is the average temperature of the study in Kelvin.
C	is the measured initial impurity content or response of interest.
N_1	is the fitted exponent on the humidity term.
N_2	is used for power or diffusion model types to fit the nonlinear impact of time on degradation rate.

There are four unique model forms resulting from the above table, as types B and D have the same model form, just

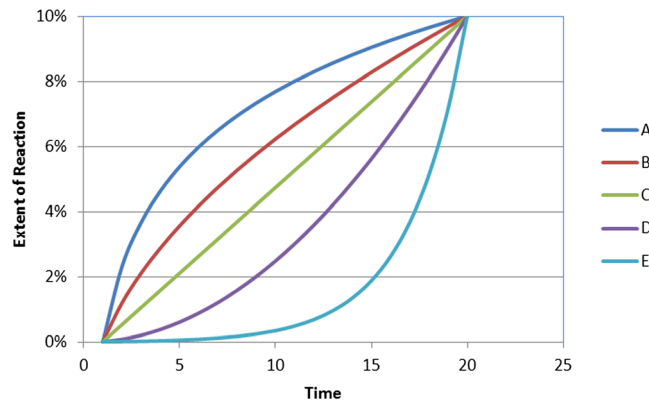


Fig. 1. Typical curve shape associated with each kinetic model type

Table I. Kinetic Equations Available in Our Accelerated Stability Modeling Platform

Model type	Description	Typical application	Nonlinear model (integrated form) used for fitting	Limitations	Terms needed to fit
A	Reaction slows exponentially with time	Reaction rate proportional to a limiting reagent that is close to being consumed	$\alpha = C + K_3 \cdot \left(1 - \exp\left(-K_1 \cdot \exp\left(\frac{-K_2}{T+273.15} + \frac{K_2}{T_{ref}}\right) \cdot H^{N_1} \cdot \Delta t\right)\right)$		4
B	Reaction slows down with time	Diffusion	$\alpha = C + K_1 \cdot \exp\left(\frac{-K_2}{T+273.15} + \frac{K_2}{T_{ref}}\right) \cdot H^{N_1} \cdot \Delta t^{N_2}$	$N_2 < 1$	4
C	Reaction is linear with time	Zero-order reactions	$\alpha = C + K_1 \cdot \exp\left(\frac{-K_2}{T+273.15} + \frac{K_2}{T_{ref}}\right) \cdot H^{N_1} \cdot \Delta t$		3
D	Reaction speeds up with time	Power law induction reactions	$\alpha = C + K_1 \cdot \exp\left(\frac{-K_2}{T+273.15} + \frac{K_2}{T_{ref}}\right) \cdot H^{N_1} \cdot \Delta t^{N_2}$	$N_2 > 1$	4
E	Reaction speeds up exponentially with time	Nucleation or reaction rate dependent on product concentration	$\alpha = \exp(C) \cdot \exp\left(K_1 \cdot \exp\left(\frac{-K_2}{T+273.15} + \frac{K_2}{T_{ref}}\right) H^{N_1} \cdot \Delta t\right)$		3

different values for N_2 . Each of the four model forms is evaluated using both RH and vapor pressure, resulting in eight models tested for fit *versus* the experimental data.

If the average impurity content at time zero (C) is well known, then this value can be directly entered into each equation being fitted. This has benefit of reducing the number of estimated model parameters by one. If C is not measured initially, then it will have to be fit, and the number of fitted parameters for each equation shown above will need to be increased by one.

A simplifying assumption that is used to help with reducing the number and complexity of models considered to five shapes is that we are primarily interested in the first 1% to 2% for impurity formation or 5% to 10% loss in drug content before the product is declared off specification. Therefore, we are not generally going to be interested in kinetics that are “sigmoidal” that initially speed up, then slow down again as the reactants are consumed. These are called “Avarami-Erofeev” reactions and will need to be transformed as $Y^* = \ln((\alpha)/(1-\alpha))$ prior to analysis, and Y^* then fit to a kinetic equation. Other rare cases may also need a transform to linearize with respect to time if they do not fit within the five curve types shown in Fig. 1. A discussion of alternate transforms to linearize with respect to time is available elsewhere (6). An example of linearizing using the Avarami-Erofeev transformation is shown later in this paper.

Nonlinear fitting techniques can be performed in a variety of commercially available software platforms such as SAS or JMP (20) to determine the kinetic model parameters for these equations. Both JMP’s nonlinear fitting routine and SAS’s Proc Nlin have been employed for this purpose within GSK. A grid search methodology is required to improve model convergence and the likelihood that one will obtain a model parameter fit that is globally optimized to result in the best possible fit and lowest model root-mean-squared error (RMSE). If no grid is used, or too coarse a grid, then local solutions could possibly be found that might model the data well, but may not be the best possible model that can be fit to the data.

Once all eight possible kinetic models are fit, then a heuristic needs to be applied in order to evaluate and rank each model to enable selection of the accepted model.

KINETIC MODEL SELECTION

Selection of the appropriate kinetic model should be done considering both the model’s applicable statistics, as well as the scientific feasibility of the calculated kinetic model parameters. Model selection is often based on comparing the model’s adjusted R^2 value or Bayesian information criteria (BIC). A potential pitfall of purely depending on statistical assessment is that it is possible to erroneously converge kinetic models with high goodness of fit that may have underlying problems with parameter inflation, and do not fall within normal limits of what is viewed as scientifically possible.

From the point of view of an engineer or chemist, the desire is for a kinetic model that fits well, but also within limits of what is scientifically considered acceptable. A model is likely to be rejected if the estimated activation energy (K_2) or level of humidity sensitivity (N_1) is outside limits on what is considered normal for a kinetic equation.

Defining “Normal” for Activation Energy and Humidity Sensitivity

To evaluate what is normal for a reaction’s activation energy, results were compiled from information provided by the Chemical Kinetics Information Center, part of the U.S. National Bureau of Standards and stored in the NIST database where activation energy divided by R was calculated for small molecule organic reactions (21). This will be referred to as the NIST dataset.

At the time of this writing, kinetic models for 537 impurities or physical characteristics from 106 accelerated stability experiments have been documented into a database and made available for this summary. This represents approximately 80% of the accelerated stability experimentation at GSK to date and covers models for impurities as well as physical phenomenon such as solvate loss, dissolution changes, hydrate formations, free base formation, and crystal form change. The resulting distribution for the K_2 (E_a/R) value can be compared between datasets in Fig. 2.

GSK’s results for K_2 (E_a/R) have the same approximate range, and mean, as the distribution from the NIST dataset,

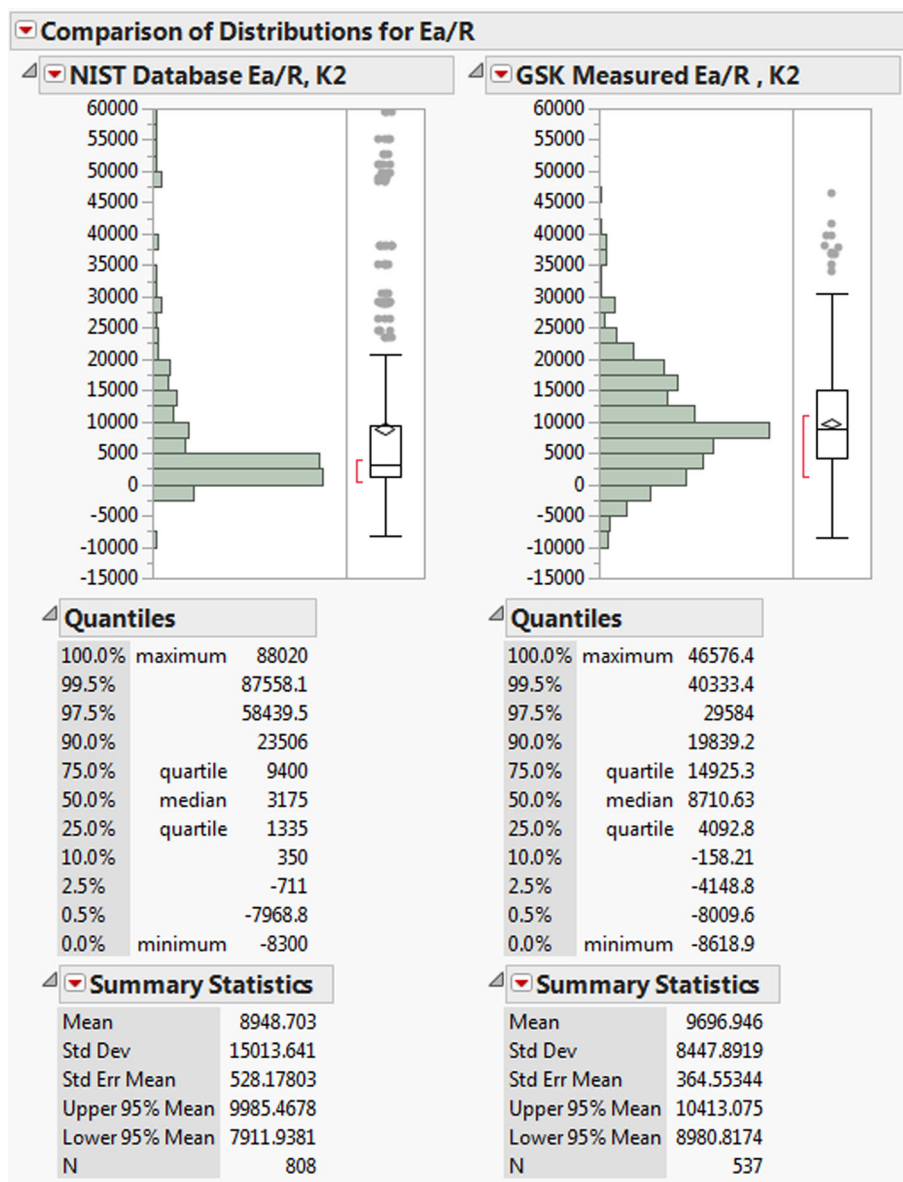


Fig. 2. Distribution of E_a/R for GSK studies compared to all organic reactions in NIST dataset

but GSK's distribution is tighter, with a smaller standard deviation. We observe a higher median value for K_2 in GSK's dataset, indicating a skew to slightly higher activation energies observed in the majority of GSK's results. We conclude that the activation energy distribution we are observing in all the studies compares reasonably well with the wide variety of organic reactions and mechanisms listed in the NIST dataset. We would expect reactions to have a K_2 value between approximately $-10,000$ and $60,000 \text{ K}^{-1}$.

Figure 3 shows the distribution of N_1 , the humidity exponent, within the 537 kinetic models. Greater than 80% of reactions have exponents between -3 and 3 indicating typical behavior as expected by order of reaction in kinetics. It must be noted however that particularly for RH-based models, the relationship between RH and water concentration in the material is not necessarily linear. There may be a cubic or higher order relationship between water concentration and RH, then a high order relationship between water concentration and the reaction

kinetics. This is likely why there is a long tail in this data all the way up to $N_1 = 8.5$ as the most extremely humidity-sensitive material observed. It is also useful to observe that slightly more than 25% of reactions have a negative humidity exponent, indicating a faster degradation under desiccated conditions.

Of the 537 kinetic models, 155 were power models with an exponent on the time term (N_2). The distribution of N_2 in GSK data (Fig. 3) shows two distinct populations. The first and largest population is centered between 0.33 and 0.66 as would be expected with solid-state diffusion reactions, indicating that diffusion of a reactant into the API crystal is the likely limiting step in the reaction. The second smaller population is centered at a value of approximately 1.4 and likely associated with induction type reactions.

It is also useful to plot the distribution of the humidity power (N_1) versus the model type. We would hypothesize that RH-driven models will show broader potential exponent values, whereas vapor pressure-driven kinetics should have

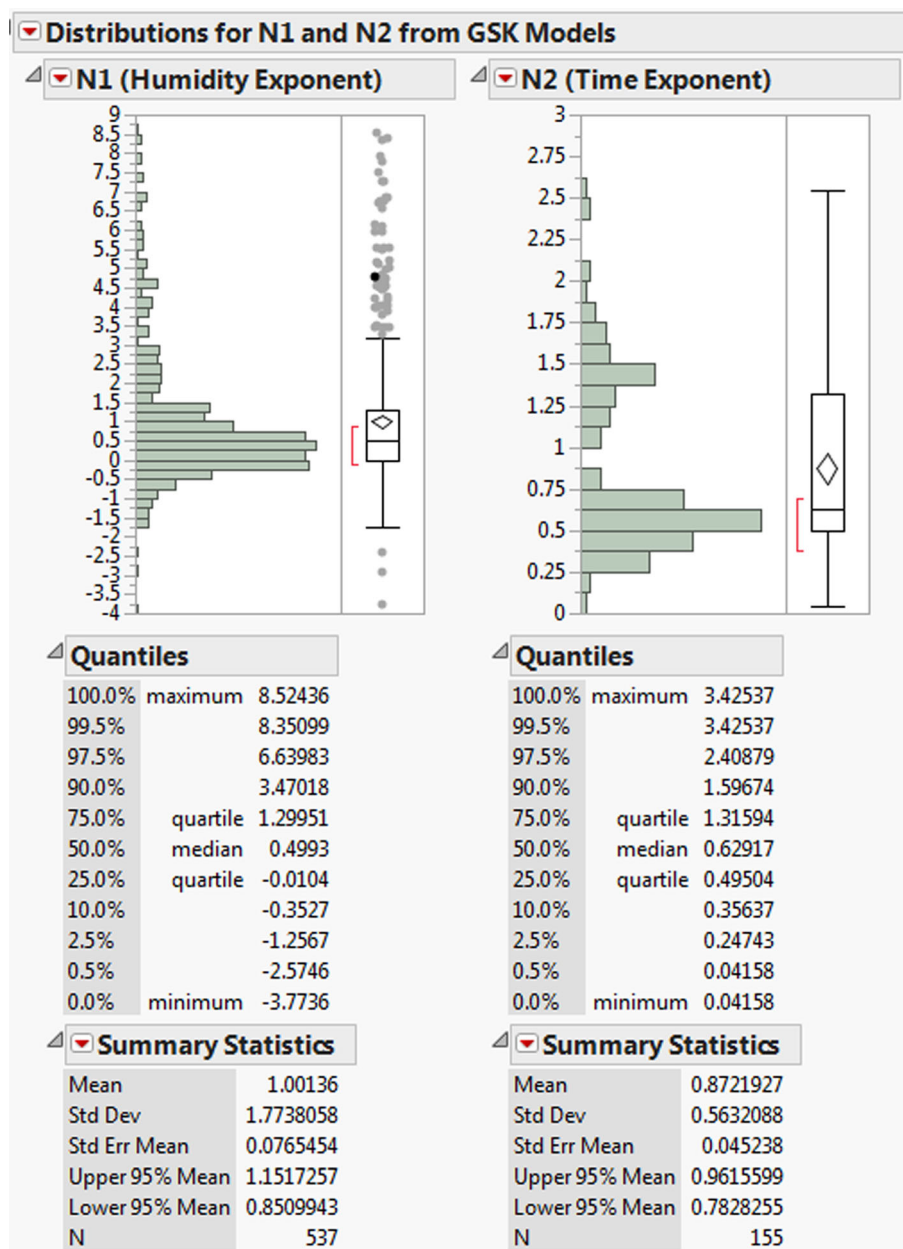


Fig. 3. Distribution of humidity exponent (N_1) and time exponent (N_2) from GSK models

smaller values. This is because vapor pressure is a true measure of water concentration in the atmosphere, while RH is a surrogate for water concentration or activity which can be correlated with RH in a highly nonlinear way. As can be seen in Fig. 4, models where RH was selected have a much wider distribution of exponent values on the humidity term. The vapor pressure-driven models are mostly distributed between 3 and -3 as would be expected from normal kinetic order, with a few extreme observations of N_1 up to 5, mostly associated with extremely high or low values of activation energy indicating unusual chemistry or phenomena.

Figure 5 summarizes which model equation type was found to provide best fit for changes in these 537 impurities or physical characteristics. If there was no capability to discriminate between RH and VP versions of model forms, the occurrence

rate of VP and RH versions of models should be approximately the same, distributed like flips of a coin. Instead, what is observed are significant differences in the numbers observed for each model form. This indicates that the system is operating as intended, with capability to discriminate between VP and RH versions of models. For instance, the RH power model occurred 93 times out of 155 observations of the power model. Using this data, a test of the null hypothesis that a random 50/50 chance is at play has a p value of 0.013; so most likely what is being observed is a tendency for power models to be driven by liquid water. This makes mechanistic sense as the majority of power models have N_2 values between 0.33 and 0.66 as would be expected for the diffusion of liquid water into a solid.

The decelerating model (type A) is a model form developed to describe a reaction rate with a limiting reagent,

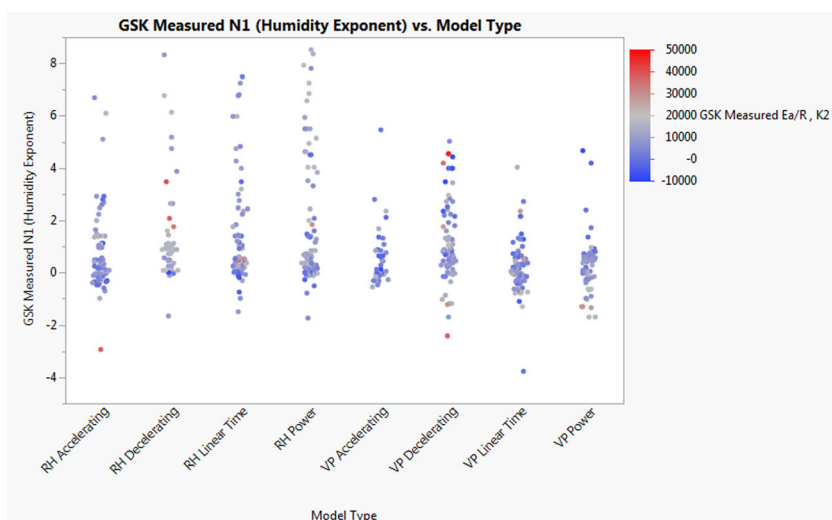


Fig. 4. Comparison of humidity exponent (N_1) values observed by model type, colored by activation energy divided by R

for example the formation of a degradation impurity due to reaction between API and a low-level impurity present in a drug product excipient. From the model type data shown in Fig. 5, vapor pressure typically drives these reactions, rather than liquid water. The VP version of this model was observed 81 times in 131 observations. Using this data, a test of the null hypothesis that a random 50/50 chance is at play has a p value of 0.0068. This indicates that these limiting reagent reactions are typically surface driven and occur with fruitful molecular encounter between water vapor, limiting reagent, and API.

The distinction between liquid water and water vapor-driven kinetics can be successfully made, it most likely does not occur due to random chance, and the observations make mechanistic sense.

Kinetic Model Selection Heuristic and an Example

The information about what parameter values are typically expected in a stability study can be used as part of the screening

process during the model selection process. A heuristic can be used for automated model selection composed of three steps:

1. Fit all possible models using RH and VP in each model form.
2. Eliminate those models that result in K_2 and N_1 values outside of reasonable scientific range.
3. Select from the remaining models on basis of the model with the lowest BIC score.

To simulate and illustrate how this works, let us start with an example theoretical dataset generated from a known kinetic equation where the rate is proportional to the square root of time (type B), RH dependent, with a K_2 value of 7500 and humidity exponent (N_1) value of 3. A fit of all possible models to the resulting dataset is summarized in Fig. 6.

The first obvious thing to note is that nearly all model forms can be forced to fit the data reasonably well, with RMSE values ranging from 0 to near 0.35 for models that

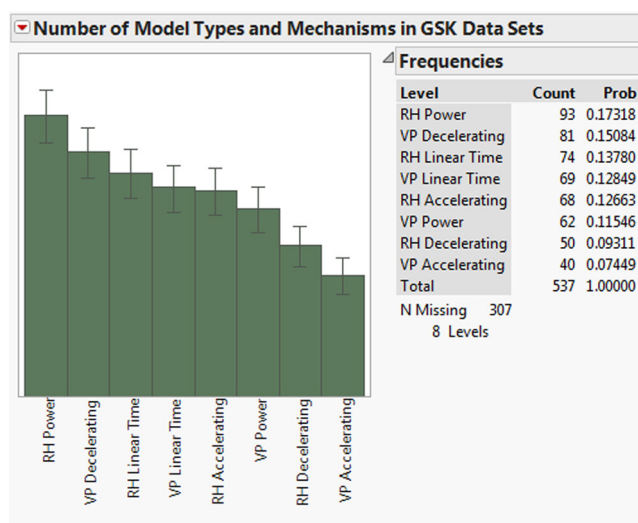


Fig. 5. Mechanisms and model types fit to GSK accelerated stability studies

Model Type	K1	K2	K3	N1	N2	nParams	RMSE	Score	AICc	BIC
RH Linear Time Kinetic Model	0.69	8374	-	3.34	-	4	0.1115	-	-22.964	-21.742
AH Linear Time Kinetic Model	118	-8801	-	3.34	-	4	0.1115	-	-22.966	-21.744
RH Accelerating Kinetic Model	4.4	-17419	-	11.19	-	4	0.3514	-	25.237	26.459
AH Accelerating Kinetic Model	1	1000	-	1	-	4	1.3584	-	82.024	83.247
RH Decelerating Kinetic Model	0.33	12601	3.7	3.35	-	5	0.0221	-	-88.206	-87.939
AH Decelerating Kinetic Model	57.33	-4588	3.7	3.34	-	5	0.0221	-	-88.3	-88.032
RH Power Model	1.9	7500	-	3	0.5	5	0	-	-848.163	-847.896
AH Power Model	193.48	-7937	-	3	0.5	5	0.0001	-	-312.928	-312.66

Fig. 6. Fit of all possible models to dataset created by an RH model, with K_2 value of 7500 and N_2 value of 3.0, as applied to a typical accelerated stability study

converged. The RH and AH (vapor pressure) versions of the models will often be close competing models. In this case, the selection of the appropriate model is clear from the lowest BIC value, but it is a situation with no noise in the data. With noise in the data, there is often a need for the scientific selection criteria.

Models with unusual values for K_2 or N_1 are automatically flagged in the software, and if the values are outside the ranges shown in Figs. 2 and 3, the model is eliminated from further consideration. Once the scientifically dubious models are eliminated, then the model with the lowest BIC could be selected.

There are cases where multiple models have similar activation energy and BIC. This can be the case where the humidity sensitivity is near zero, in which case it does not matter whether the RH or VP version is selected as they are both going to be equivalent.

DESIGN OF THE INITIAL EXPERIMENT FOR ACCELERATED STABILITY

Before starting the experimental design for accelerated stability, it is highly recommended to carry out a DSC or TGA study on the components in the formulation to ensure no phase changes occur within the temperature and humidity ranges proposed for the study. If a phase change occurs, this will likely result in models that cannot be extrapolated to lower temperatures. Similarly, there may also be significant concerns if solid-state form changes occur within the proposed temperature

ranges of the study as this may impact the reaction kinetics due to changes in surface area, reactivity, or solubility. In addition, a study relating relative humidity to moisture content of the material being tested, referred to as the moisture sorption isotherm, should be performed since this data will be utilized later when simulating the degradation of the test material under long-term storage conditions and when packaged.

A good experimental design is often associated with little or no correlation among the factors being studied. However, design optimization is also a function of the model being used. For a given number of design points, an optimal design will seek to minimize some function of the variances of the unknown model parameter estimates. For example, it is possible to set up an experiment where temperature and RH are varied independently of each other in a hexagonal fashion, similar to Fig. 7.

From experience, a hexagon has been noted as a useful initial design as it provides at least four levels of humidity, which allows observation if there is a marked change in mechanism at the highest or lowest humidity values. In a few rare occurrences, materials have been noted to slow down or speed up more than would be predicted indicating potentially a change in mechanism at extremes of RH.

Since relative humidity, water vapor pressure, and temperature share a three-way relationship, it is not practical or indeed possible to develop a study that is completely uncorrelated in every possible parameter that might be needed in a kinetic model.

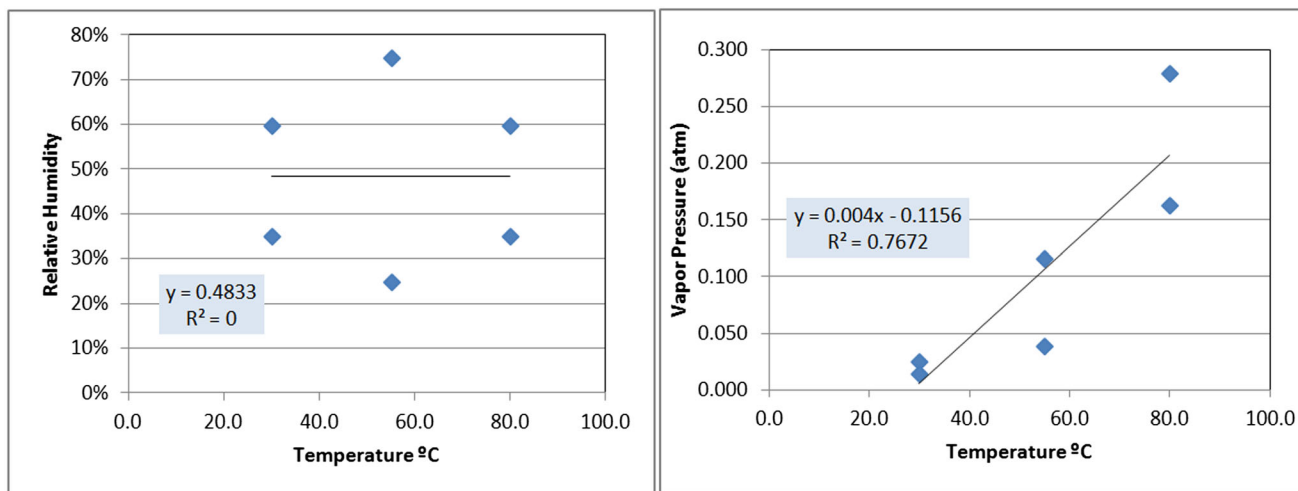


Fig. 7. A designed experiment with no correlation of RH and temperature (left) can lead to strong correlation of vapor pressure versus temperature (right)

If the above experiment was conducted, and the rate of degradation reaction or transformation was dependent on the vapor pressure, the effects of temperature and vapor pressure would be significantly confounded due to correlation. Due to the (parametrically) nonlinear form of the kinetic model, there may be no way to statistically remove the parameter correlation effects associated with temperature from the increase in water vapor pressure.

The aim of an initial designed accelerated stability experiment should be to minimize the correlation between temperature and RH and between temperature and vapor pressure so that either possibility can be explored during initial model development without too much confounding due to correlation.

Another factor to consider in experimental design is dealing with the conflict that arises from the spread of temperature in the study. A large temperature spread is desirable as it helps reduce the error of estimation for activation energy. However, a large temperature range makes it impossible to design an experiment of limited data points where temperature and vapor pressure are not strongly correlated. For practical purposes, the recommendation is to limit the spread of temperature in the study to 30°C or perhaps 35°C unless the mechanism for degradation is already known. Most of GSK's accelerated stability studies have a 20°C spread and range from 50°C to 70°C, or 60° to 80°C.

To help accomplish an accelerated stability experimental design, a spreadsheet was developed in Microsoft Excel where the user can enter the maximum and minimum study temperatures and humidities, and then a Solver routine seeks to create a design that minimizes the sum squared correlation between (T,RH) and (T,VP). The resulting experimental temperatures, humidity, and the corresponding, commercially available saturated salt solutions are then displayed for the user. The resulting output for the recommended accelerated stability study ranging from 50°C to 70°C is shown in Fig. 8.

Once the first accelerated stability study has been completed for a given product, the nature of the mechanism and the nonlinear model form is then assumed known. It

might then be desirable to design a new study that is optimized with minimum experimental points possible.

Further design enhancement is possible using D-optimal design methodology (22). In addition, Monte Carlo studies can be used to assess the effectiveness of both the configuration of the design points and the total number of experiments. The design configuration and total number of experiments run will have a direct effect upon the size of any prediction interval for shelf life. Monte Carlo studies can also assess the ability of various experimental designs and sample sizes to be able to discriminate between models.

Clearly, a significant concern with numerically optimized studies that tend to feature only two humidity levels and three temperature levels is if the kinetics were actually more complex. If the initial screening experiment found that the points at 0% or 80% humidity were poorly predicted, there is a significant risk that running, say, a standard factorial DoE design at only extreme humidity levels will result in models that poorly fit more moderate humidity levels. These designs would not be able to detect this. In addition, traditional factorial designs are not necessarily optimal for (parametrically) nonlinear models. The advantage of the initial hexagonal screening design described in this section is that it has more humidity levels to be able to detect problems like these.

A D-optimal experimental design should only be used if there is certainty of a continuous single model over the humidity range being studied and strong prior information on the values of the model parameters. If the certainty is high, then D-optimal designs can be useful to obtain model parameter estimates with the minimal data points possible.

ESTIMATION OF UNCERTAINTY IN KINETIC MODELS—BOOTSTRAPPING

For the nonlinear regression models used, the statistical inferences reported by the JMP and SAS software

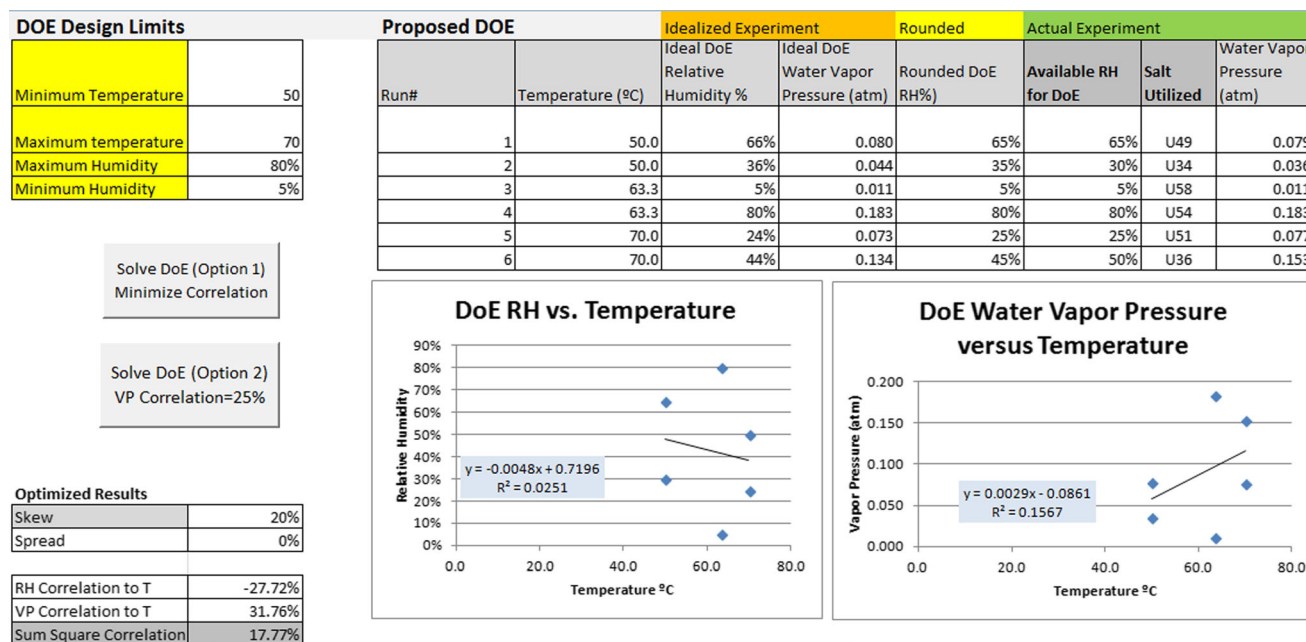


Fig. 8. Screening experiment design tool in Excel showing the default initial experiment

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are based upon large-sample asymptotic theory. This means that the confidence intervals and p values assume that there is a sufficiently large amount of information for all of the (unknown) model parameters. Given that accelerated stability designs may not always contain enough data to justify the large-sample theory underlying the inferences reported by JMP or SAS, an alternative analysis is needed. The alternative analysis used here is “parametric bootstrapping” (23). Parametric bootstrapping uses Monte Carlo simulations from the fitted model to assess the uncertainty of the model parameter estimates. Unlike the standard methodology in JMP or SAS, parametric bootstrapping (or “bootstrapping” for short) gives good estimates of model parameter estimate uncertainty even for moderate sample sizes. Bootstrapping is also useful in that if the large-sample results from JMP and the bootstrap results agree, then this provides some evidence that the experimental design configuration and overall sample size provide a good amount of information about the unknown model parameters.

The bootstrapping procedure produces a joint distribution that approximates the underlying joint distribution of all of the model parameter estimates. This joint distribution can be used to create bootstrap-based confidence intervals and p values for the model parameter estimates. This joint distribution can also be used to create confidence intervals and prediction intervals associated with the nonlinear regression

model at various values of the input variables (experimental factors).

The bootstrapping procedure will then need to be run “ N ” number of times to obtain a confidence interval. An example of what this procedure looks like as N is increased from 50 to 1000 bootstraps for a type “A” kinetic equation is shown in Fig. 9.

Cross sections of the bootstrap distribution are then taken to give predictions of attribute levels at different times. Quantile distribution analysis is then done on the bootstrap error distribution at each time point to estimate the probability of passing specification in Fig. 10. This approach enables project teams to make risk-based decisions regarding product stability with a quantitative assessment of the risk of failing to meet product specifications at any particular day of interest.

EXPERIMENTAL METHODOLOGY

To conduct an accelerated stability study, samples are stored in glass vials exposed to conditions of predetermined temperature and humidity. Individual small chambers are prepared with separate temperature and humidity probes continually logging the environments in the chambers. The chambers utilize saturated salt solutions to maintain the humidity in the environment and then are placed in an oven to control the temperature.

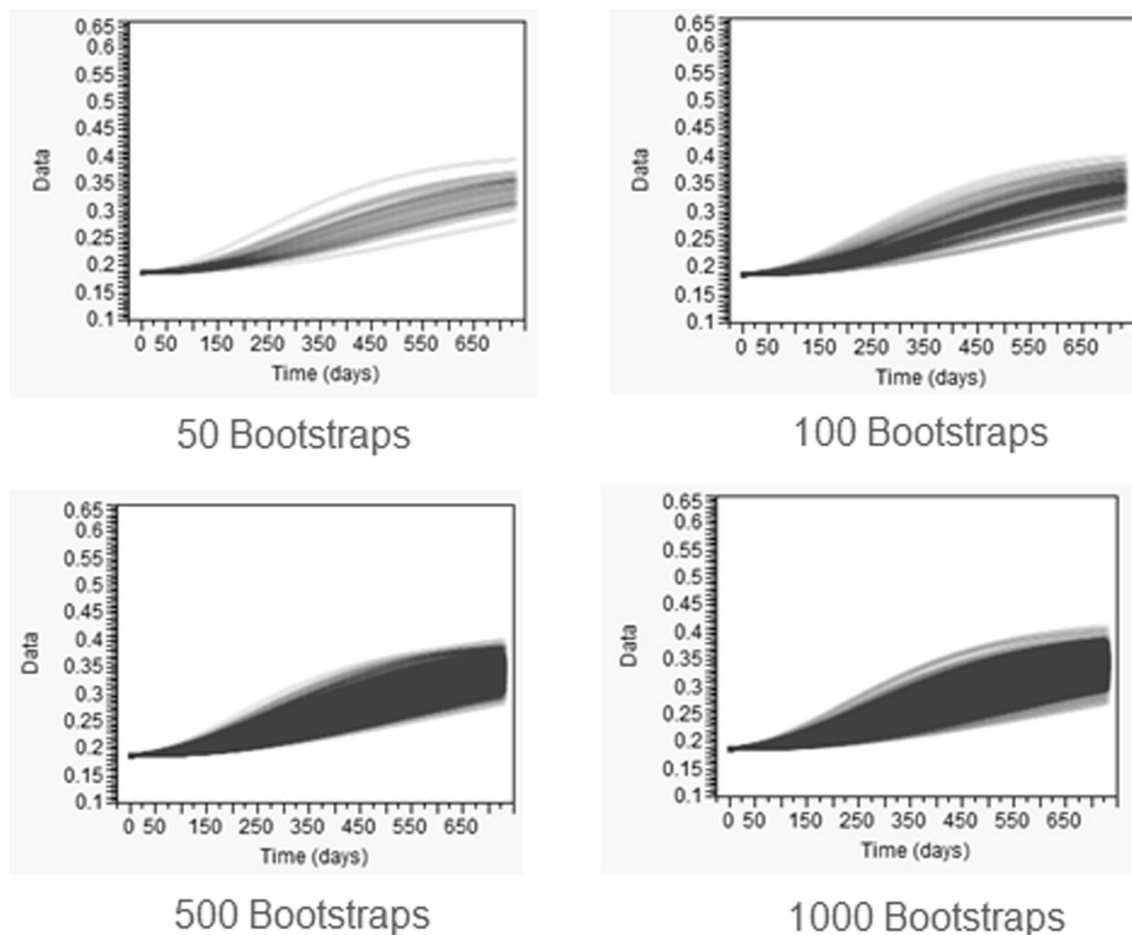


Fig. 9. Impact of increasing number of bootstraps

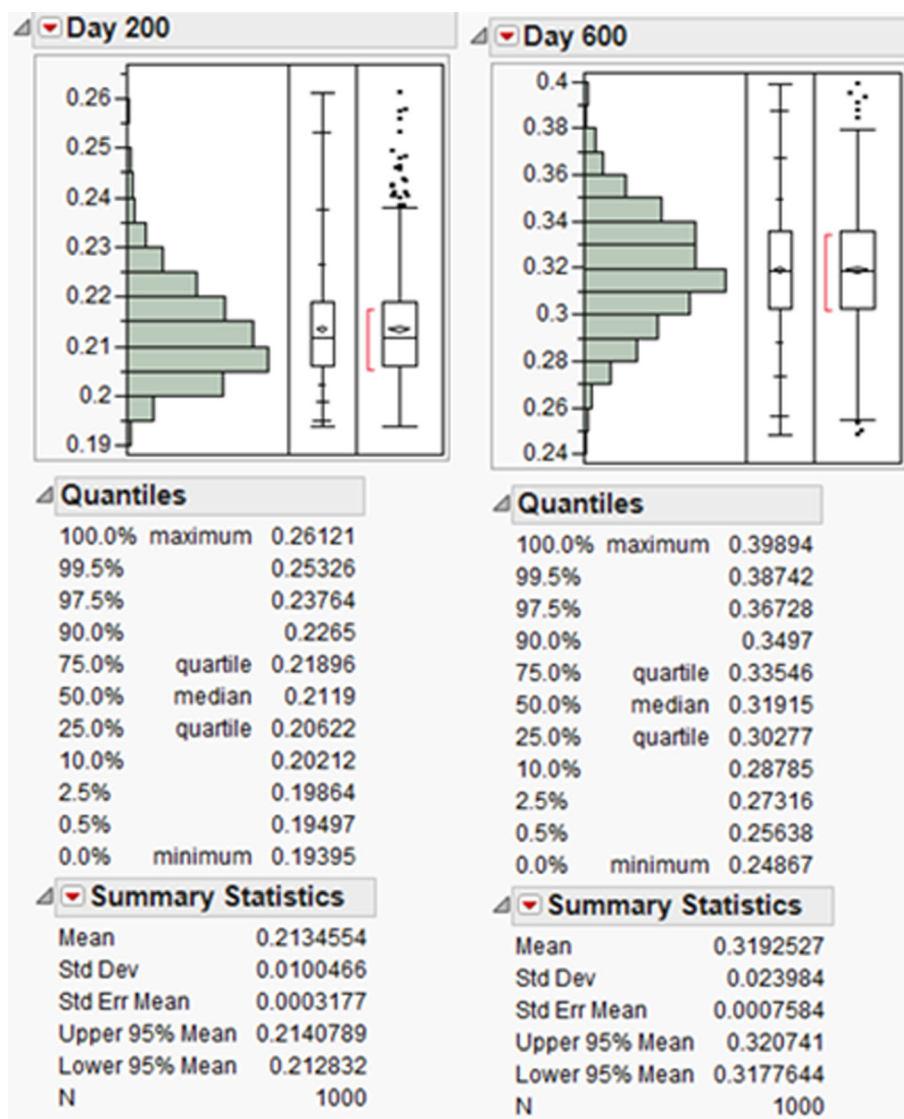


Fig. 10. Quantile distribution analysis of bootstrap predictive distribution at varying times

Table II. Drug Product 81 Accelerated Stability Experimental Data

Temp (°C)	RH (%)	Time (days)	Impurity level (parts per 10 ⁴)					
			Imp101	Imp966	Imp545	Imp799	Imp283	Imp102
71.3	77.6	4	6.7	14.8	9.3	7.7	38.2	13.2
79.3	27.7	4	8.3	13.5	9.5	6.9	36.8	14.0
61.8	39.9	7	7.4	6.0	9.4	6.1	34.1	13.1
71.5	51.3	7	7.5	12.9	9.7	7.2	35.6	13.7
71.3	77.6	7	7.2	13.2	8.8	8.8	37.8	14.1
79.3	27.7	7	9.8	16.3	9.9	7.5	37.7	14.0
61.9	6.6	10	9.2	5.7	9.4	6.0	34.9	13.8
61.8	39.9	10	8.1	6.9	9.4	6.4	34.6	14.1
71.5	51.3	10	8.2	14.6	9.7	7.8	36.1	13.4
50.4	74.8	14	6.8	6.0	9.5	6.6	34.4	13.7
61.9	6.6	14	10.1	5.5	9.6	6.5	34.6	14.5
50.4	74.8	10	6.9	5.4	9.4	6.6	34.3	14.3
61.8	39.9	0	5.2	2.0	9.5	5.8	29.8	13.1
Spec. limit (parts per 10 ⁴)			35	35	35	34	60	80

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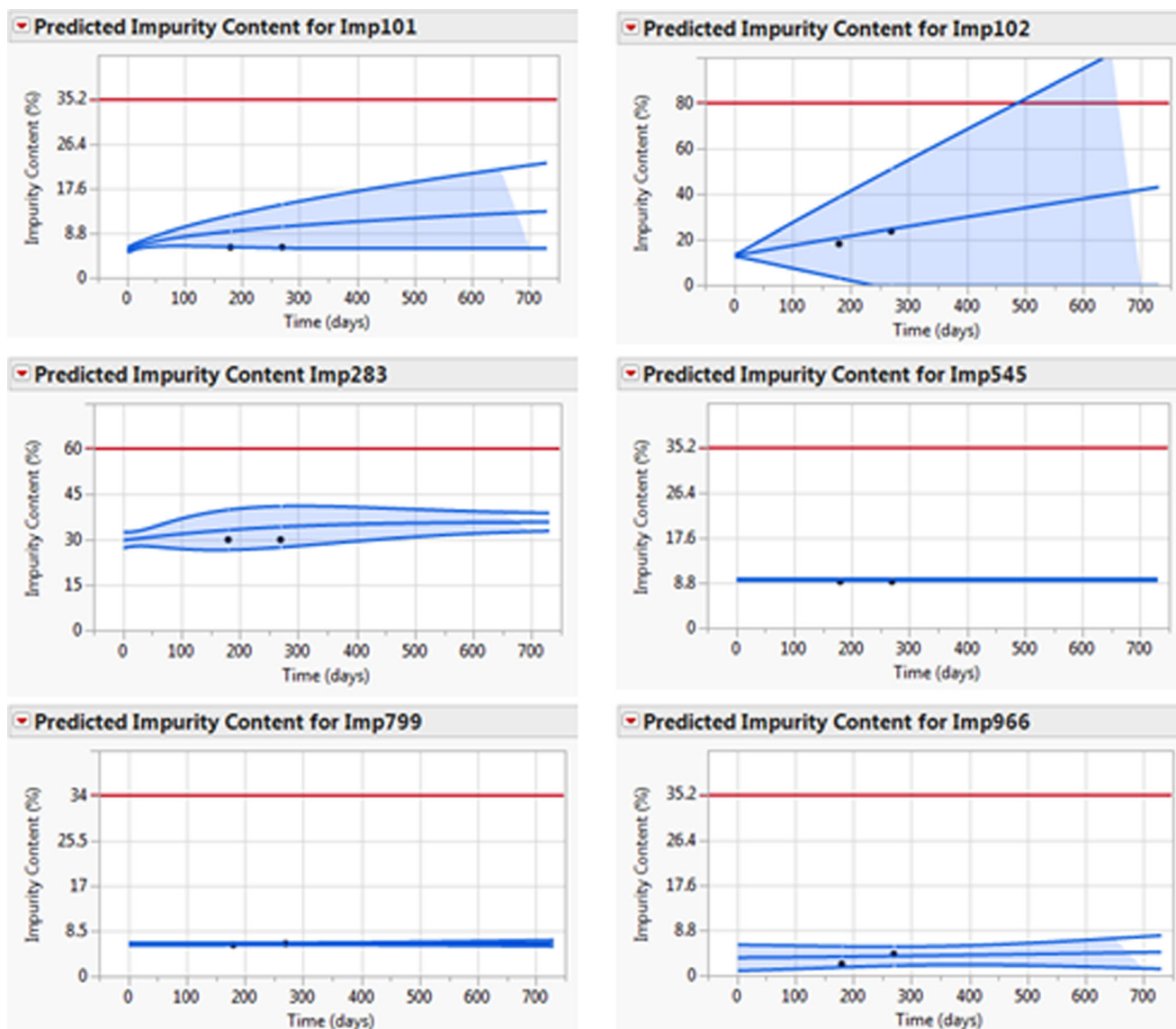


Fig. 11. Drug product 81 impurity models overlaid with long-term data at 30°C, 65% RH in HDPE bottle with desiccant. Impurity levels measured in parts per 10,000

Samples are preweighed into glass vials to enable efficient analysis of the samples produced during the study, stored uncapped (and therefore exposed to the environment), and removed from the chambers at the predetermined time

points. Samples are then re-equilibrated to room temperature and humidity, capped, and stored refrigerated until the end of the study when they are tested altogether in order to reduce influences from analytical variability.

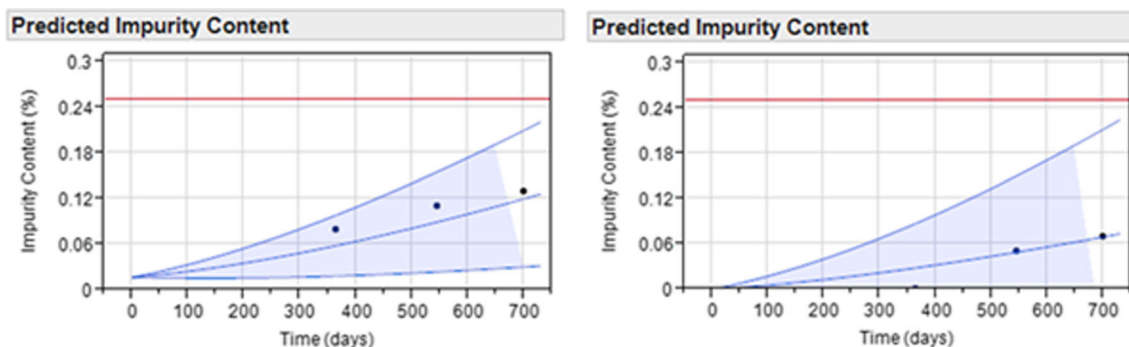


Fig. 12. Two impurities for drug product P, models overlaid with long-term data

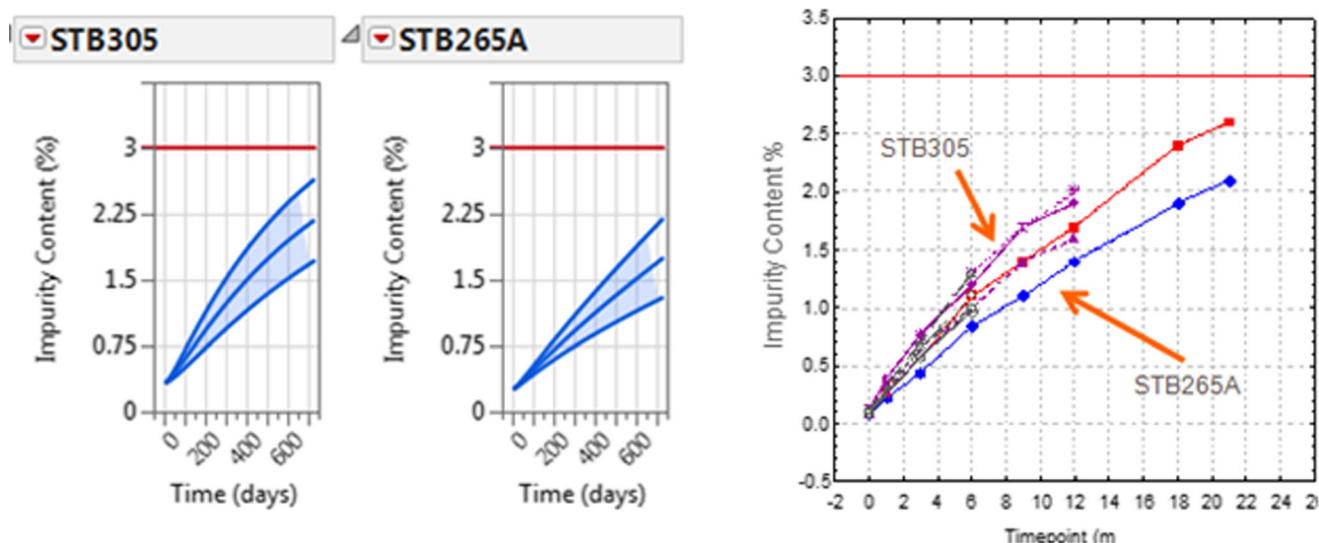


Fig. 13. Drug product S batches 305 and 265A in accelerated stability prediction (*left and center*) compared to long-term observation 2 years later (*right*)

Samples from the study are then analyzed using compound-specific methodology, *e.g.*, by high pressure liquid chromatography

(HPLC) and/or gas chromatography (GC) in order to quantify degradation products which had been formed during storage.

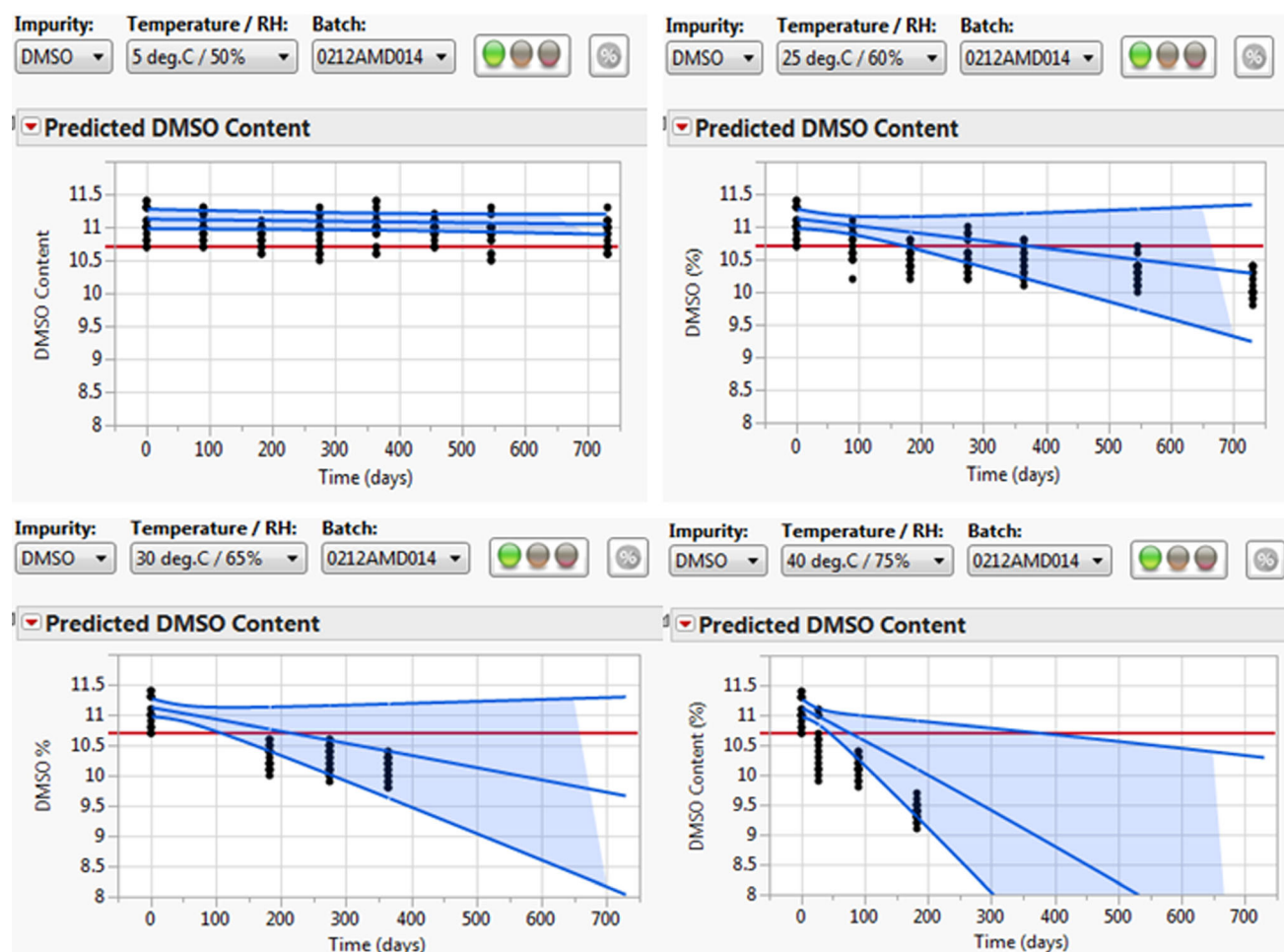


Fig. 14. Predicted long-term stability (*solid lines*) and actual stability data (*circles*) at 5°C/ambient RH (*top left*), 25°C/60% RH (*top right*), 30°C/65% RH (*bottom left*), and 40°C/75% RH (*bottom right*) boundaries indicate upper and lower confidence interval of model predicted mean

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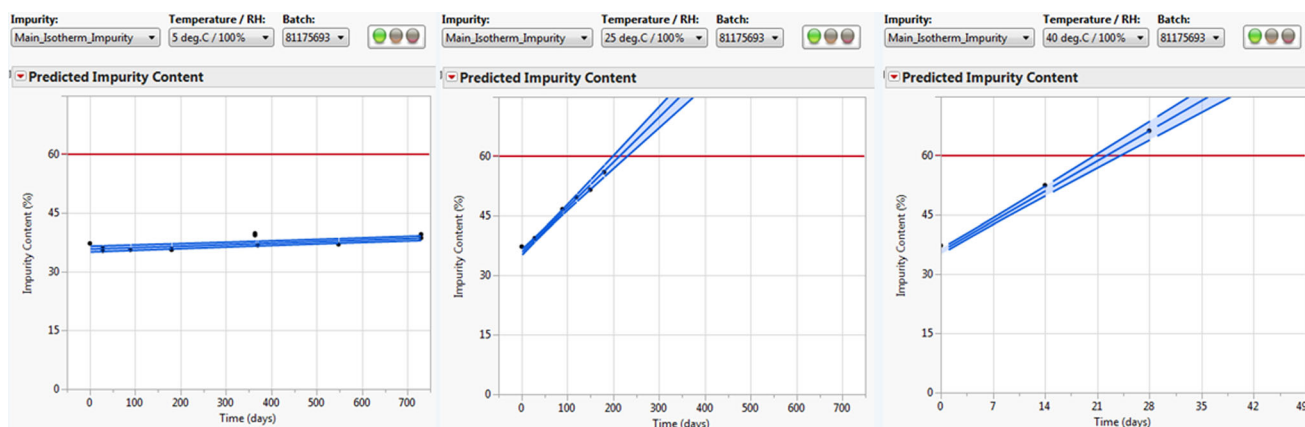


Fig. 15. Biologic product main isotherm impurity data at 25°C (*left*) to 40°C (*mid*) and 5°C (*right*) follows a simple Arrhenius model, allowing accelerated experiments to predict stability under refrigerated conditions

Where possible, data is reported to 4 decimal places to enable maximum statistical analysis, and reported to the limit of quantification of the method on the day of analysis (typically 0.02% a/a). Where a degradant is detected but below the limit of quantification, the value is omitted from analysis. Where there was no degradant detected, a value of “0” is utilized for the modeling.

A control sample for each study is stored in the same format as the other samples, capped and refrigerated for the duration of the study, and tested concurrently with the other study samples and used as the “initial” sample for the study.

COMPARISON OF PREDICTED LONG-TERM STABILITY TO MEASURED RESULTS

In a couple of instances, “control” studies were set down at the same time as accelerated stability studies to compare the predictions with long-term stability data. The comparison to long-term data is not a finite measure of success, however, since an enhanced understanding of stability and improved datasets influencing project decisions do not require exact overlay with

long-term data and it is unlikely that this type of data where the same batches were used for accelerated and long-term stability will be regularly available. Most long-term studies involve at the very least a change in batch from the original accelerated studies, and the starting impurity content may be different.

Prediction of long-term stability needs to utilize a model describing the permeation of moisture into the packaging. The degradation kinetics found from a short-term study are integrated over time with the long-term humidity prediction to create a long-term prediction for impurity content.

In the first example demonstrated below (Table II and Fig. 11), accelerated stability studies were carried out on two batches, one batch from an older synthetic route (B) and one from a new route (C). Model predictions for the old route (B) were compared with long-term stability data in order to verify the suitability of the models generated for the new route (C), and to justify the decisions made based on the model predictions. The resulting accelerated stability data for the older manufacturing route is shown in Table II and the developed models are shown in Fig. 11 with the long-term data available for the older synthetic route.

Table III. Data from Accelerated Stability Experiment for Drug Product “N”

Temperature (°C)	RH (%)	Time (days)	Assay (% w/w)	Conversion (%)
50	50	0	100.3	0.7
50	50	4.8	96.8	4.2
50	50	9.8	93	8
50	50	13.7	90.8	10.2
59	30	4.8	97.5	3.5
59	30	9.8	98.3	2.7
59	30	13.7	97.3	3.7
70	12	4.8	97.5	3.5
70	12	9.8	98.8	2.2
70	12	13.7	97	4
71	74	2.8	37.8	63.2
71	74	6.9	10.5	90.5
86	8	4.8	97.5	3.5
86	8	9.8	98.3	2.7
86	8	13.7	98.5	2.5
85	53	2.8	72.5	28.5
85	53	6.9	12.5	88.5
Specification limit (% w/w)			91.5	9.5%

Table IV. Data Extrapolations to Find Isoconversion Time in ASAP Prime Software for Drug Product “N”

Temperature (°C)	Humidity (% RH)	Isoconversion time (days)	Std dev (days)
50	50	14.6	1.2
59	30	56.4	18.4
70	12	55.0	17.5
71	74	0.3	0.1
85	53	1.0	0.0
86	8	95.6	44.8

If one looks at the data in Table II, it is hard to intuitively observe that Imp102 would actually be the main impurity to be concerned with. Imp101 had a much stronger response in the accelerated stability study, but the kinetic modeling showed that Imp102 had a lower activation energy, would slow down less at lower temperatures, and would be the main impurity to be concerned about in long-term stability. This illustrates the general problem with other approaches such as the isoconversion approach. Kinetics can observe and predict this behavior within several weeks to a month, whereas an isoconversion approach would have to wait a very long time to observe failure.

In another example, a 2-week accelerated stability experiment was conducted for drug product “P” and the resulting predictions were then overlaid with the 12-, 18-, and 24-month long-term stability data (Fig. 12).

Accelerated stability studies have been used as part of a DoE to assess differences in stability in potential formulations. In this following example for “drug product S,” two batches prepared at the extremes of the formulation composition as part of a DoE were set down in a 2-week accelerated stability study prior to setting down samples for long-term stability for all the batches in the DoE. The 2-week accelerated stability study correctly predicted the significant difference that would be observed between these two batches and provided reasonable prediction of the long-term trends that were then available 2 years later as shown in Fig. 13.

The long-term predictions from the short-term accelerated stability experiment can also be made about some types of physical phenomena. One example is an API that has been isolated as the corresponding dimethylsulfoxide (DMSO) solvate resulting in a need to understand the rate of DMSO loss from the crystal structure. The DMSO loss was examined in

a short-term, 2-week accelerated stability study at temperatures ranging from 50°C to 70°C and relative humidities ranging from 5% to 80%. A model was fitted and then compared to actual long-term stability data developed from the NDA and confirmatory studies for this drug product. The long-term data available includes 769 observations of DMSO content for product in HDPE bottles with desiccant. Data simulated from the accelerated stability kinetic model integrated with a model of the HDPE bottle with desiccant is shown to closely correspond with long-term data as shown in Fig. 14.

It is important to bear in mind with physical phenomenon that additional mass transfer effects may be present besides the kinetics of DMSO leaving the crystalline structure as measured during the accelerated stability study. There is a more rapid drop of free DMSO content leaving the system within the first month in the long-term data which is particularly evident at higher temperatures. The free DMSO content is not bound in the crystal structure and accumulates in the time period from manufacture to start of the stability study within the excipients in the tablet. Since the free initial DMSO content was higher in the final stability batches compared to the accelerated stability batch, a slightly faster DMSO loss is observed in the first month. The data tracks downwards after the first month in a similar manner as predicted by the kinetics observed in accelerated stability.

In addition, one also needs to consider other mass transfer effects that might be operating in these systems. Since the accelerated stability experimentation provides a large sink in the form of a hydrated salt solution for the DMSO to exit the API matrix, what is predicted is the rate of desolvation under sink conditions. If pure API is wrapped in a foil container where DMSO cannot escape, then little to no rate of DMSO loss is observed from the crystal structure. Since tablets of this pharmaceutical are primarily composed of diluents, and the storage bottle includes a desiccant, significant locations are available for the DMSO to sink, and thus, API tested under sink conditions behaves similarly to API in tablets in sink conditions.

It was not envisioned that the same model and software development could also be used for chemical or physical degradation of biologic products stored as solutions in vials, but the modeling for those has typically shown to be successful. The accelerated experiments range in temperature from 25°C to 40°C, and humidity is a constant 100% RH within the sealed vials. Since RH is constant in the experiments, the humidity sensitivity term (N_1) reduces to

Table V. Resulting Parameter Fits and Predicted Shelf Life from ASAP Prime Software for Drug Product “N” with Comparison to Measured Shelf Life

ln A	18.7 ± 3.1	
E _a	15.6 ± 2.0 kcal/mol (65.3 ± 8.4 kJ/mol)	
B	0.103 ± 0.008	
R ²	1.00	
Condition	Measured shelf life (months)	Predicted shelf life with PVdC (months) ^a
25°C/60% RH	17.9 (13.4–32.5)	11.2 (9.2–13.6)
30°C/65% RH	8.9 (5.7–11.3)	6.2 (5.3–7.3)
30°C/75% RH	5.3 (4.4–7.6)	5.2 (4.5–6.1)
40°C/75% RH	1.3 (0.8–1.3)	2.0 (1.8–2.2)

^a Shelf life predicted by the ASAP Prime software

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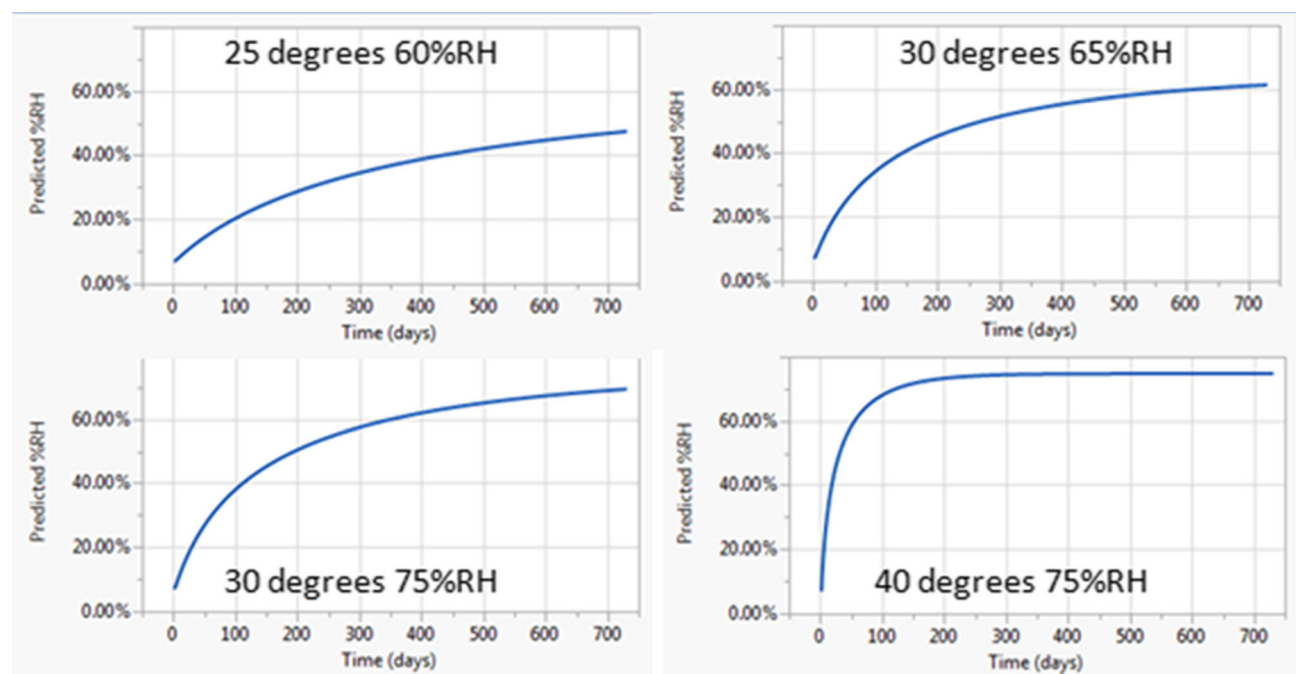


Fig. 16. Predicted RH values over time for PVdC blisters utilized for drug product “N”

zero, and what is fitted is the remaining portions of the kinetic models covering the impacts of temperature and time. The accelerated experiment can be used to predict the main isotherm impurity and caliper reduced impurity formation for various products under refrigerated conditions at 5°C, as shown in Fig. 15.

An important caveat for biologic samples in frozen state is that the equations developed from samples in liquid state often cannot predict the degradation rate of frozen products. The kinetics could be slower due to reduced molecular mobility, or faster than predicted, potentially due to the heterogeneous freezing of ice,

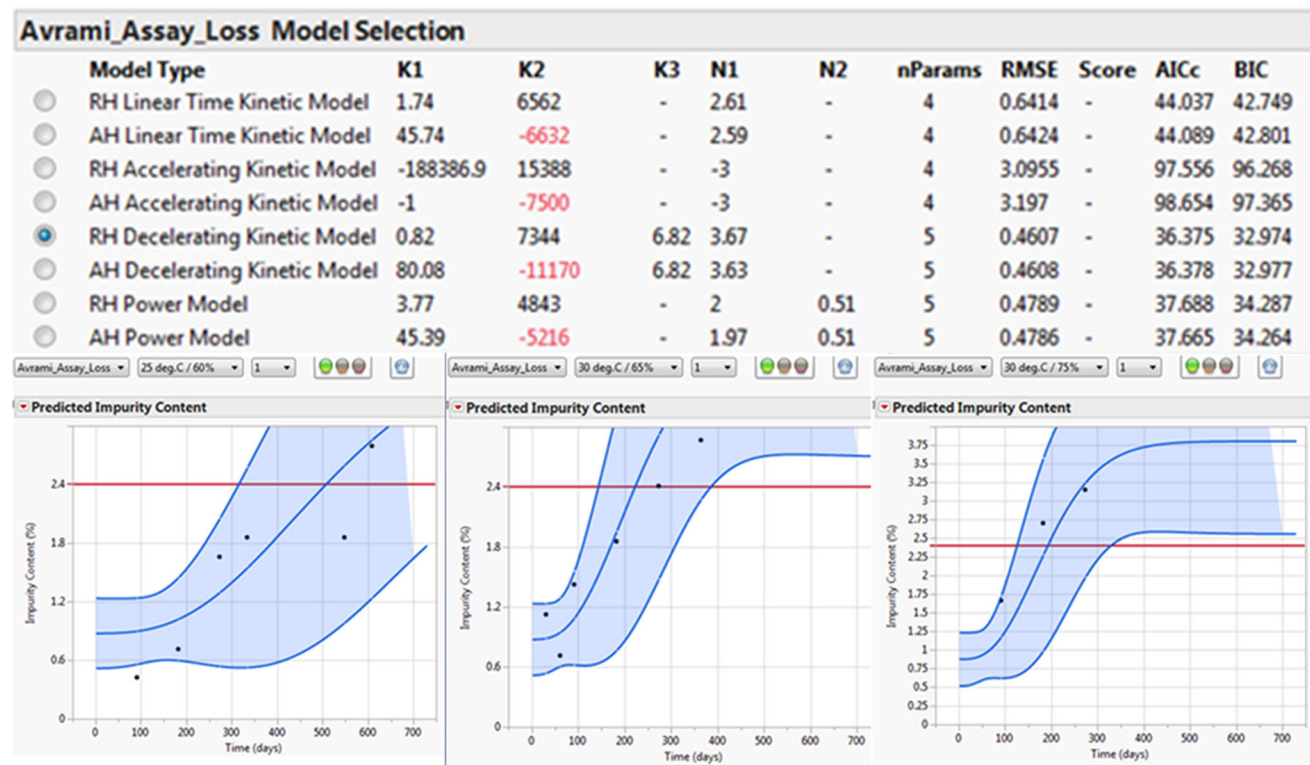


Fig. 17. Kinetic model prediction and the model's predicted conversion extent (utilizing Avrami transform) for drug product “N” compared to actual long-term data

Table VI. Comparison of Observed Time to Failure *versus* That Predicted by ASAP Prime and GSK's Methodology for Drug Product "N"

Storage condition	Observed time to failure (months)	Predicted time to failure (months) ^a	Predicted time to failure (months) ^b
25°C/60% RH	17.9	11.2 (−37.4%)	16.8 (−6.3%)
30°C/65% RH	8.9	6.2 (−30.2%)	7.4 (−16.7%)
30°C/75% RH	5.3	5.2 (−1.2%)	6.3 (18.8%)
40°C/75% RH	1.3	2.0 (52.0%)	1.5 (12.5%)
Average error		30.2%	13.5%
Slope of actual <i>versus</i> predicted data		1.84	1.10

^a Shelf life predicted by the ASAP Prime software^b Shelf life predicted using GSK methodology described in this paper

which tends to concentrate the salts in solution thus potentially driving a faster reaction rate.

A final example discussed in this paper is drug product "N," which has been modeled using both ASAP Prime, a commercially available software package that uses the isoconversion approach and assumes an exponential relationship to relative humidity, and the methodology described in this paper, enabling a direct comparison of the predictive methodologies. Drug product "N" assay decreases with time as a function of temperature and humidity. An accelerated stability experiment was done, with the resulting data shown in Table III.

The resulting data was analyzed utilizing the ASAP Prime software. The extrapolations to find the isoconversion time and the resulting calculated isoconversion times for each condition are shown in Table IV.

The isoconversion times were then fit in ASAP Prime utilizing Eq. (3). The resulting parameter fits, and the predicted time to failure is shown in Table V.

The rate of assay loss of drug product "N" is particularly humidity dependent, and it is useful to show the predicted humidity values in the PVdC blister cavity under varying conditions to allow the calculations to be replicated by others at a later date (Fig. 16).

To start GSK's analysis of the data, it was observed in these data that the assay loss follows a typical "S" curve associated with Avrami kinetics. As was mentioned in the model development, the model that is exponential with time has an implicit assumption that the extent of conversion is low, and the deceleration that occurs at high extents of conversion is well beyond the conversion being observed. Since this assumption is not true in these data, some data exceed 90% conversion, and it is appropriate to use the Avrami transform to help linearize the data prior to further analysis. To do this, a value called "conversion" was calculated as

$$\text{Conversion}\% = 101\% - \text{Assay}\%$$

The Avrami transform utilized to assist in the data linearization was then

$$' \text{Avrami Assay Loss}' = \ln \left(\frac{100 \cdot \text{Conversion}}{100 - \text{Conversion}} \right)$$

The resulting transformed data was then analyzed using the methodology described in this paper. The resulting kinetic equation selection and prediction of long-term data is shown in Fig. 17.

Table VI shows the comparison of accuracy between GSK's methodology developed in this paper *versus* that previously described in literature and utilized in ASAP Prime. Since the packaging in PVdC blisters was done at very low RH of 7%, the impact of an assumption of exponential RH behavior becomes evident. An exponential function cannot go to zero, and it can only approach a value of one as the RH substantially decreases. Therefore, a predicted reaction rate cannot approach zero, and ASAP Prime overestimates the rate of degradation. In contrast, a kinetic model that assumes water is a reactant with a reaction rate proportional to RH raised to a power is able to approach zero as the RH declines to zero, and more accurate predictions are made. The difference in analysis methodology becomes particularly evident for humidity-sensitive compounds where desiccation is used to increase the shelf life.

CONCLUSIONS

Based on our experience to date, we find that drug product stability is not typically exponentially affected by increasing RH as assumed by current practice in literature. The assumption of exponential behavior may overestimate the resulting kinetics when extrapolated to low relative humidity, and thus, the methodology described in this paper is believed to be more accurate. The typical curvature with respect to humidity found in GSK studies is closer to cubed root or square root rather than exponential with power terms typically between −1 and 1 and the majority of data falling within −2 and 2 as might be typically expected for the order of reaction in reaction kinetics.

We have chosen to develop an approach where the degradation kinetics are modeled directly rather than focus on the isoconversion approach as it allows us to improve the logistics of doing a large number of ASM studies, and makes greater use of all the available degradant *versus* time data rather than reduce this to a single value. We have chosen to focus on four kinetic model types each with a different curvature with respect to time, and for each model type, a decision is made whether the kinetics are driven by direct interaction with water in the atmosphere (vapor pressure driven), or by liquid water (RH driven). To make this decision, eight models are fit for each dataset and then compared. Model selection is based on two criteria: (1) scientific and then (2) statistical assessment of the quality of fit. First models are eliminated which have model parameters that are viewed as scientifically infeasible, based upon acceptable ranges for activation energy (K_2) and order of reaction (N_1), as defined from literature and current findings in

GSK studies. Once the scientifically improbable models are eliminated, the best model is chosen based upon the lowest BIC score.

The aspiration of this model development would be to predict long-term data ranging up to 2 years from a 2- to 4-week accelerated stability study. The approach we have developed has thus far been applied to more than 120 development projects with more than 600 impurities, and has so far shown good fit with long-term stability trends.

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